TRADEGAUNTLET: TESTING THE LIMITS OF TRADING COMPLIANCE CONTROLS

ADAPTIVE EVASION, ORDER SPLITTING, AND CANCELLATION TIMING

PREPRINT

Sandip Nandi
Independent Researcher

ABSTRACT

Pre-trade gates and surveillance rules are usually evaluated by counting alerts. We argue that three quantities must be measured separately on matched market paths: whether a rule alerts, whether a gate removes the gain from moving the price, and how much legitimate execution the gate blocks. TradeGauntlet, a replayable limit-order-book simulator, makes this possible. A detector that shares a threshold with a gate is silent on gated traffic by construction. With a price response fitted to Nasdaq order-book data for five stocks, a seeded search over split-order schedules, scored on held-out paths, finds schedules that evade the implemented conduct rules under every control we test, also when only depth near the touch moves the price. An aggregate cap at twice reference depth removes between none and about half of their gain. An order-to-trade ratio and a learned detector together catch every searched schedule but flag 23% of a legitimate large maker's sessions. Legitimate makers also gain from the price response, and the cancellation throttle, not the cap, carries most of their cost. A paired check in ABIDES finds no detectable gain from displayed depth. With two LLMs trading legitimate tasks, every alert was a self-match by a market maker, which the gates remove by construction; the informative effects are the gates' cost in fills and a prompt reminder that reduces Sonnet's self-matching. All trading is simulated.

Keywords trading compliance · market surveillance · spoofing · limit order book simulation · adversarial evaluation

1 Introduction

A gate may reject a 60-lot order while accepting six 10-lot orders at the same price. Checking each order separately misses the combined display. A cap on total resting quantity addresses that gap, but its usefulness depends on the market response and the legitimate trading it prevents.

This paper asks how to evaluate such controls against a trader who adapts to them. Pre-trade gates and surveillance rules are usually assessed separately and by counting alerts, but an adaptive trader can search for the gap between them, and a gate can silence a detector without removing the gain the detector was meant to catch. The question has become more pressing now that brokers let automated agents place and cancel orders on a customer's behalf [23, 24], but it applies to any adaptive trader. Most of our experiments use scripted and searched strategies; Section 10 adds two language-model agents given legitimate tasks.

We built TradeGauntlet for this study: a small limit-order-book simulator coupled to a workflow for fixing scenarios, comparing controls on matched paths, and replaying the resulting orders. We distinguish whether a surveillance rule flags a sequence, whether a gate removes its gain from the simulated price response, and how much legitimate execution the gate prevents. An alert records a pattern; a rejection records an intervention. Neither alone tells us whether the intervention removed the gain or whether the trading was lawful.

Our contributions are:

1. An evaluation method for stacks of trading controls that scores alerts, removed gains, and legitimate execution separately on matched market paths, with exact replay of every episode. Two simple observations show when

such an evaluation is uninformative by construction: a detector that shares a threshold with a gate cannot alert on gated traffic (Observation 1), and a step response has a closed-form activation boundary in the cap multiple (Observation 2).

2. An adaptive adversary: a seeded search over split-order schedules, selected on training paths and scored on held-out paths with bootstrap confidence intervals, which replaces hand-written counterexamples, together with two baseline detectors and an independent check in ABIDES (Section 6).

3. A threshold-free price response fitted to Nasdaq order-book data for five stocks, which turns the question of whether a cap protects into a measured quantity rather than an assumed one, with robustness checks on the response window, the unit mapping, and search restarts.

We also report costs to legitimate traders from the cancellation throttle and a detector-window counterexample on ordinary market making. The simulator makes these interactions inspectable; it does not estimate how often they occur in markets. Finally, we run Claude Haiku 5.5 and Sonnet 5.5 as traders under the same controls on held-out market paths (Section 10).

2 Related work

Wang and Wellman [1] study spoofing with trading agents, and Wang et al. [7] extend the analysis to strategic responses and market-design changes. Cartea, Jaimungal, and Wang [18] model the trade-off between spoofing gains and expected fines. Lee, Eom, and Park [34] measure the strategies and profits of spoofing traders in Korean market data, and Williams and Skrzypacz [35] show when spoofing arises in equilibrium. Wang, Vorobeychik, and Wellman [36] propose cloaking part of the order book, an intervention that, like our caps, limits how much displayed depth can move prices. These studies explain economic mechanisms that our fixed scripts and imposed price response take as given.

Tao et al. [19] derive spoofing conditions from a multilevel imbalance model, calibrate it to TMX Level-2 data, and propose monitoring based on distributional distance. Chen, Hsieh, and Yang [20] use Taiwan derivatives order and trade data to study potential spoofing and a price-banding intervention. Both bring market evidence to questions of detection or trading restrictions. We fit our response model to Nasdaq data for five stocks (Section 5); the rest of the comparisons measure effects within the simulator.

Two working papers address evidence that this design lacks. Do and Putniņš [13] evaluate indicators using prosecuted cases and out-of-sample tests. Fabre and Challet [10] use Level-3 data to estimate price responses and possible spoofing gains. External labels and an estimated response would strengthen our evaluation; varying an assumed threshold does not supply either.

Learning traders can adapt to detection and enforcement. Byrd [2] studies an RL trader that discovers spoofing and a learned recognizer that discourages it. Wang and Wellman [3] model adaptation between manipulation and detection, while Cartea, Chang, and García-Arenas [12] examine manipulation by learning algorithms. Gu et al. [17] vary liquidity and spoofing strategies, and ABIDES [4] supports richer agent interaction and communication latency. Tuccella, Nadler, and Şerban [37] train a recurrent detector for spoofing on order-book data. Realism metrics for limit-order-book simulators [38] and generative market simulators [39] address a question we leave open: whether our minimal book reproduces market stylized facts. Our adversary is simpler than these learners: it searches a fixed family of split-order schedules rather than learning a policy, which keeps each found strategy inspectable and replayable.

Pandey and Rajwal [25] evaluate LLM judgments about potential market abuse using variants of enforcement-case narratives. RISCTrade [26] evaluates adherence to risk mandates during trading across multiple markets. We focus on what happens at the order level when a gate or surveillance rule changes, including the cost to legitimate execution.

InvestorBench [27] compares LLM agents across financial tasks and assets. FinAgent [28] develops a multimodal trading agent with tools and memory. QFinZero [29] provides standardized financial tools, stateful brokerage simulation, order-lifecycle management, and replay. Here, we compare interacting controls on matched market paths, with scripted, searched, and LLM traders.

ReguSim [8] separates a trader's reasoning, attempted orders, enforcement, and monitoring evidence; its reported environment uses exogenous prices and fully executes accepted orders. Borjigin and He [9] propose constrained execution with a runtime shield, leaving the full cryptographic proof pipeline for future work. Kurshan, Balch, and Byrd [5] discuss agent-based governance, and Byrd [11] studies trading with LLM-generated social posts. Our LLM traders choose orders from the book alone, without a social-information channel.

3 TradeGauntlet: simulator, gates, and detectors

3.1 Market mechanics

TradeGauntlet uses a custom matching engine, implemented for this project, with integer prices, price/time priority, and partial fills. It is not built on ABIDES. Scenario runners fix the configuration and random seeds, execute paired control conditions, record attempted and accepted actions, and replay saved attempts to check event hashes and outcomes. Cap multiples, response thresholds, throttle windows, session lengths, and quoting cadences can be varied through these runners. Orders execute within the simulator, with no broker or exchange connection.

Component	Specification
Book and matching	One evaluated account and one instrument; integer ticks, price/time priority, partial fills
Timing	One second per attempted action; background replenishment every second; default decisions every 60 seconds, with exact re-quote times in the cadence boundary checks
Background liquidity	Three price levels per side, 10 lots per level, centered around the fundamental plus any response
Exogenous path	Fundamental starts at 1,000; increments -1, 0, +1 with probabilities 1/4, 1/2, 1/4 every 30 seconds; noise quantities 0, 1, 2 with probabilities 1/2, 1/4, 1/4 every 10 seconds
Risk and fees	100-lot worst-case position commitment; fee 0.05 tick-lots per filled lot
Sessions	720 seconds in the core study; 720 and 3,600 seconds in the throttle tests; no warm-up

Table 1: Simulator configuration.

The fundamental and noise streams are generated before trading starts. For a given seed and session length, every gate and response condition therefore receives the same exogenous draws. Unfilled market quantity expires. Background quotes are replaced between successive subject actions, so splitting an order also changes how long the display remains in the book.

Immediately before each replenishment, let B and A be total remaining buy and sell quantity, including the subject. The imbalance is $I = (B - A)/\max(1, B + A)$. In the responsive condition, background quotes shift by two ticks when I exceeds a threshold t, by minus two ticks when I is below $-t$, and otherwise do not shift. The unresponsive condition omits this shift. The original threshold is 0.4; the sensitivity grid varies it. This step response is stipulated, not estimated from market data.

Because every result under the step response hinges on an unestimated threshold, we add a second, threshold-free response. In the *linear* condition, background quotes move κ ticks in the direction of I with probability $|I|$, and otherwise stay centred on the fundamental, so the expected displacement is κI ticks. This follows the empirical regularity that the probability of the next price move rising increases with order-book imbalance, roughly linearly for large-tick stocks [31], and that short-horizon price changes are approximately linear in order-flow imbalance [30]. We use $\kappa = 1$ as a stress setting. The uniform draws that decide each move are generated before trading from a separate seeded stream, so all gate conditions on a seed share them. Section 5 fits the move probability to market data.

3.2 Pre-trade gates

All conditions enforce the position-commitment bound. The legacy gate additionally rejects a limit submission above twice external reference depth, more than two resting price levels on a side, and a predicted self-match. Reference depth is the median external quantity at the two touches, excluding subject orders. The aggregate gate adds a per-side cap: remaining subject limit quantity plus the full proposed quantity must not exceed m times reference depth. Originally $m = 2$. The sensitivity grid varies this aggregate multiple while retaining the legacy per-order multiple of 2.

Both gates can throttle new limit and market submissions when $C/N > 0.9$ and $N \geq 20$, where C and N count subject cancellations and accepted submissions in the trailing window. The original window is 1,800 seconds. Rejected attempts do not increase N; cancellations remain available. Throttle-off conditions make the minimum submission count unreachable, without changing other gate checks. The aggregate check runs at submission and does not rebalance existing orders as external depth changes. It also checks a marketable limit at its full submitted quantity, so its rejection cost is intentionally conservative.

3.3 Pattern definitions

For a canceled order i, let q_i be submitted quantity, d_i its submission-time external reference depth, and a_i its age at cancellation. Let h_i be the fraction of q_i filled before cancellation and z_i its distance from the same-side touch at submission. For a subject fill f on the opposite side, define $Q(i, f)$ to hold when i was nonmarketable at submission,

submission precedes f, f precedes cancellation, $a_i \leq W$, and $h_i < 0.1$. Event sequence numbers determine ordering within a timestamp. Table 2 lists the implemented rules.

Signal	Condition for an alert
Individual pattern	Some i and f satisfy $Q(i, f)$, $q_i \geq 3d_i$, and $z_i \leq 3$ ticks.
Layer pattern	For one f and one displayed side, qualifying $Q(i, f)$ orders occupy at least three distinct prices. No individual-size or distance test is added.
Aggregate pattern	For one f and displayed side, at least two $Q(i, f)$ orders have $z_i \leq 3$ and positive remaining quantity $r_i(f)$. Their sum is at least $3 \max_i d_i$, three times the largest reference depth in that group.
Self-match	The subject is both buyer and seller in a fill.

Table 2: Conduct-alert patterns. Their union is the conduct-alert signal.

Remaining quantity $r_i(f)$ is measured immediately before the shared fill, after any earlier partial executions. The conduct-alert union contains these four signals; a separate cancellation-ratio diagnostic is excluded. The current scoring default is $W = 180$ seconds, shared by the individual, layer, and aggregate patterns. Table 3 retains the original 120-second configuration to document the initial split-order comparison; the cadence grid varies W explicitly. These patterns do not implement a complete legal test or estimate trader intent.

3.4 Comparisons

Terminal wealth V is cash plus inventory valued at the terminal fundamental, after fees. For gate g, parameters p, and seed s, the response-dependent difference is

$$D(g, p, s) = V_{\text{resp}}(g, p, s) - V_{\text{unresp}}(g, p, s).$$

The two conditions use the same gate and exogenous path. Wealth and D are measured in tick-lots, one tick of price on one lot. Positive D indicates a benefit from this particular response toggle; it is not a legal or general welfare measure, and it is not the profit of the strategy: a schedule can have $D > 0$ and still lose money overall. Executed lots, rejected actions by reason, and response activations are reported separately.

The split probe alternates sides. Each cycle submits six 10-lot orders at one price, executes two lots on the opposite side, and cancels the surviving display. The legitimate controls are small and large two-sided makers, a patient maker, a passive scheduled executor, and a 10-lot slice executor. We assign these scripts their objectives; the assignments have not been independently reviewed. The scheduled executor is named `vwap_executor` in the code, but uses neither a volume forecast nor a VWAP benchmark.

4 Gate–detector coupling

Gate	Original alerts	With aggregate signal	Executed lots	Wealth
off	0/40	40/40	24.00 [24.00, 24.00]	22.05 [14.80, 32.80]
legacy	0/40	40/40	24.00 [24.00, 24.00]	22.05 [14.80, 32.80]
aggregate	0/40	0/40	24.00 [24.00, 24.00]	-25.95 [-33.20, -15.20]

Table 3: Six-by-ten split probe, `pilot.review_study`, seeds 3000-3039, responsive market. Both gates include the original throttle. The 120-second scoring window is retained from this comparison; rescoring at the current 180-second default leaves these alert counts unchanged. Alerts count episodes; volume and wealth are mean [minimum, maximum]. Wealth is in tick-lots.

Each split order passes the individual-size check, and all occupy one price level. The individual and layer detectors flag none of these 40 paths; the aggregate detector flags the admitted sequence. Under the aggregate gate, however, the absence of an aggregate alert is not evidence about the detector. It follows from the shared inequality.

Observation 1 (Gate–detector coupling)**.** *Suppose every order in a candidate group was admitted by the aggregate gate with multiple m, and the aggregate pattern alerts when the group's remaining quantity at the shared fill is at least $k \max_i d_i$. If $m < k$, the aggregate pattern never alerts on that group.*

Proof. Let j be the last group order admitted. At its admission the gate requires that the remaining same-side quantity, including q_j, be at most $md_j \leq m \max_i d_i$. Orders admitted before j can only lose quantity afterwards, through fills or cancellation, and no order grows after admission, so the remaining quantity at the shared fill is also at most $m \max_i d_i < k \max_i d_i$. □

With $k = 3$ and the original $m = 2$, the zero alert count under the aggregate gate is guaranteed, and says nothing about independent detector accuracy. More generally, when a gate and a detector threshold the same quantity, evaluating the detector on gated traffic measures the gate.

5 Fitting the price response

We use the five free LOBSTER samples, AAPL, AMZN, GOOG, INTC, and MSFT on 21 June 2012, reconstructed from Nasdaq messages at 10 levels (148,000 to 669,000 book updates per stock). AAPL, AMZN, and GOOG traded between about $220 and $580 with spreads of several ticks; INTC and MSFT traded near $27 and $30 and usually had a one-tick spread, so they are large-tick stocks in the sense of [31]. On a one-second grid, excluding the first and last ten minutes, we regress the mid-price change over the next h seconds, measured in units of the prevailing half-spread, on the imbalance of the top three levels, which matches the simulator's three background levels. Intervals use a block bootstrap with five-minute blocks. The simulator's half-spread is one tick. If its spread stands for the market's spread, the one-second slope is directly comparable with κs, and we set the fitted scale s to a stock's one-second point estimate. If instead a simulator tick stands for a market tick, the comparable scale is the slope times the median half-spread in ticks; Section 6 reports results under both mappings. AMZN and GOOG give slopes close to AAPL's, so the adversary uses AAPL, INTC, and MSFT as fitted settings.

	Spread		Top three levels			Best level
Stock	**(ticks)**	**Obs.**	**1 s**	**5 s**	**10 s**	**1 s**
AAPL	15	18,028	0.018 [−0.003, 0.039]	−0.083 [−0.155, −0.009]	−0.150 [−0.279, −0.016]	0.052 [0.031, 0.075]
AMZN	12	15,340	0.032 [0.015, 0.049]	0.069 [0.007, 0.129]	0.063 [−0.044, 0.166]	0.036 [0.022, 0.050]
GOOG	26	11,152	0.016 [−0.005, 0.037]	0.028 [−0.039, 0.097]	0.052 [−0.062, 0.166]	0.025 [0.007, 0.044]
INTC	1	18,889	0.348 [0.287, 0.417]	1.169 [0.932, 1.406]	1.721 [1.321, 2.130]	0.201 [0.172, 0.229]
MSFT	1	19,220	0.479 [0.406, 0.574]	1.515 [1.270, 1.803]	2.041 [1.683, 2.476]	0.252 [0.222, 0.284]

Table 4: Slope of the future mid-price change, in half-spreads, on order-book imbalance, 21 June 2012. Spread is the median quoted spread; observations are one-second grid points. Brackets are 95% block-bootstrap intervals. Slopes at other horizons for the best level are saved with the results. Produced by `paper_study.calibration`.

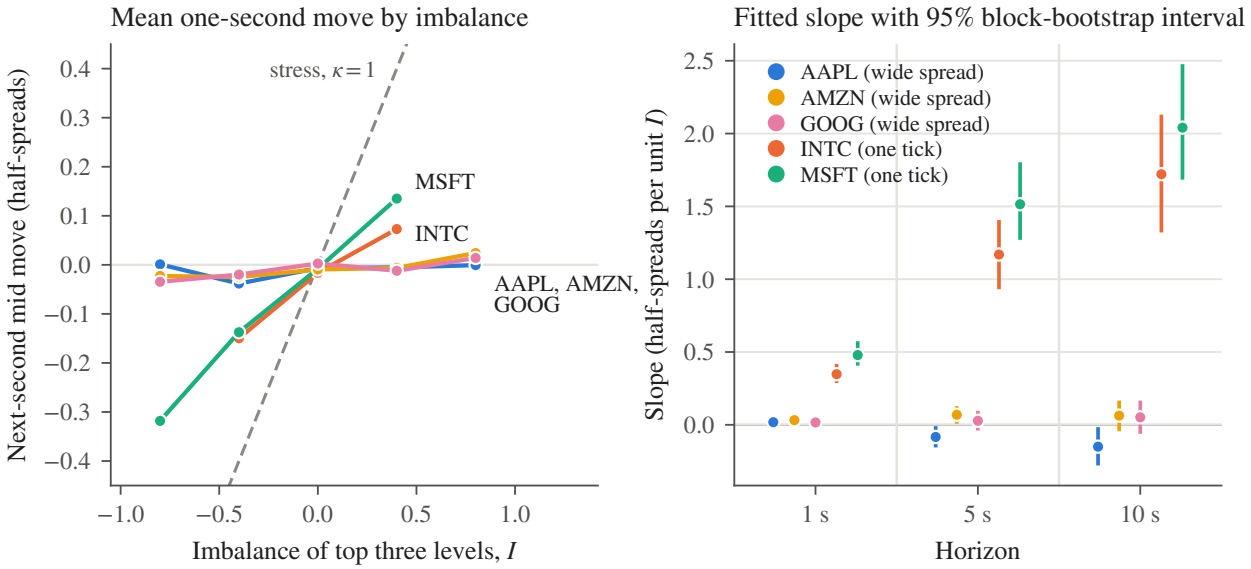


Figure 1: Fitted price response. Left: mean next-second mid move by top-three-level imbalance, in five bins, omitting bins with fewer than 30 observations; the three wide-spread stocks overlap near zero. The dashed line is the $\kappa = 1$ stress setting. Right: slope at each horizon with its 95% block-bootstrap interval.

The stocks split by tick size. For the three wide-spread stocks, the top-three-level slope is at most 0.032 half-spreads per unit of imbalance at one second and at most 0.069 at any horizon; for AAPL the sign reverses at five and ten seconds. For INTC and MSFT, imbalance predicts the next move strongly: the one-second slope is 0.35 and 0.48 half-spreads

per unit of imbalance, about a third to a half of the $\kappa = 1$ stress setting, and it grows with the horizon to about two half-spreads at ten seconds. The simulator applies its response each second, so we use the one-second slopes; the longer-horizon slopes are not comparable with a per-second response. Measured in market ticks instead, the one-second slopes are close for all five stocks, from 0.13 (AAPL) to 0.24 (MSFT) ticks per unit of imbalance, so the contrast depends on the unit. Gould and Bonart [31] study the best-level queue; for the one-tick stocks, our top-three slope is about twice the best-level slope (Table 4). These are predictive regressions on five stocks on one day; displayed orders that are not intended to trade may move prices less than organic depth, and the regressions do not identify the causal effect of one trader's orders.

6 Adaptive evasion

The results so far rest on scripts we wrote and on a response threshold we chose. This section removes both. The runner is `paper_study.adversary`; its plan, search logs, held-out episodes, and summary are saved, and a verify command reruns every held-out evaluation and reproduces the summary exactly.

Adversary. The adversary chooses a split-order schedule from the parameterized family of Section 3.4: the number of orders (2 to 8), total displayed quantity (10 to 80 lots), price spread behind the touch (0 to 5 ticks), submission gaps, lifetime overlap (full, staggered, or disjoint), and the delay before a 2-lot opposite-side execution. Each candidate is scored on 10 training paths by its mean response-dependent difference D minus 100 times its fraction of episodes with a conduct alert, so any alert-prone schedule loses. The search draws 40 random candidates and then makes 40 single-parameter local moves, keeping improvements. The selected schedule is then scored on 100 held-out paths that the search never saw. Searches run separately for each of four control settings (no gate; legacy; aggregate cap at $m = 2$; aggregate cap at $m = 1$) and six responses, with the cancellation throttle on: 24 searches in all.

Responses. Besides the step responses at $t = 0.4$ and $t = 0.2$, we use the linear response of Section 3.1 with $\kappa = 1$ as a stress setting, and three fitted settings in which the move probability is $s|I|$ with s estimated from data for one stock (Section 5).

Response	**No gate**	**Legacy**	**Aggregate,** $m = 2$	**Aggregate,** $m = 1$
Step, $t = 0.4$	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Step, $t = 0.2$	48.00 [48.00, 48.00]	48.00 [48.00, 48.00]	48.00 [48.00, 48.00]	0.00 [0.00, 0.00]
Linear, $\kappa = 1$	9.78 [9.26, 10.30]	8.24 [7.62, 8.88]	6.14 [5.60, 6.70]	3.32 [2.86, 3.80]
Fitted, MSFT	2.84 [2.42, 3.28]	3.14 [2.68, 3.60]	2.68 [2.30, 3.06]	1.37 [0.97, 1.76]
Fitted, INTC	2.00 [1.64, 2.38]	2.68 [2.26, 3.12]	2.00 [1.62, 2.38]	1.10 [0.82, 1.38]
Fitted, AAPL	0.00 [0.00, 0.00]	0.12 [0.04, 0.22]	0.04 [0.00, 0.10]	0.04 [0.00, 0.10]

Table 5: Held-out mean D, in tick-lots per session, of the schedule the search selected for each control and response, over 100 held-out paths (seeds 9100-9199). Brackets are 95% bootstrap intervals. No held-out episode in any cell raised a conduct alert. Training paths are seeds 9000-9009; the throttle is on.

Three results stand out. First, the detectors do not stop a searching adversary. With no gate and the $\kappa = 1$ response, the fixed six-by-ten probe earns 12.06 tick-lots per session (95% CI 11.38 to 12.72) but alerts on all 100 held-out paths; the searched schedule keeps 9.78, about 81% of that gain, with no alerts. It places two 40-lot orders, one and five ticks behind the touch, and cancels the first before submitting the second, so the only order live at the execution sits outside the detectors' three-tick distance condition. The same holds under the responses fitted to the one-tick stocks: the fixed probe earns 4.32 (INTC) and 5.86 (MSFT) with an alert on every path, and the search finds alert-free schedules earning 2.00 and 2.84.

Second, how much the aggregate cap protects depends on the response, and under the fitted responses it protects only partly. Under $\kappa = 1$, the cap at $m = 2$ cuts the searched gain from 9.78 to 6.14, and $m = 1$ to 3.32. Under the INTC and MSFT fits, $m = 2$ leaves the searched gain essentially unchanged in this search (2.00 and 2.68, against 2.00 and 2.84 with no gate), and $m = 1$ roughly halves it (1.10 and 1.37); the larger search below revises the MSFT comparison. The cap does remove the fixed probe's alerts, but the probe still earns 2.30 (INTC) and 3.04 (MSFT) under the cap with the throttle off, about the gain the search finds. Only the step responses produce the all-or-nothing protection of Appendix A.1, and Observation 2 predicts which step cells are zero.

Third, under the half-spread mapping of Section 5, the size of the gain follows the stock's tick size. Under the AAPL fit, which stands for the three wide-spread stocks, no searched schedule earns more than 0.12 tick-lots per session and the fixed probe earns 0.50 (0.32 to 0.70) with no gate. Under the one-tick fits, the searched gains in Table 5 are 20 to 30%

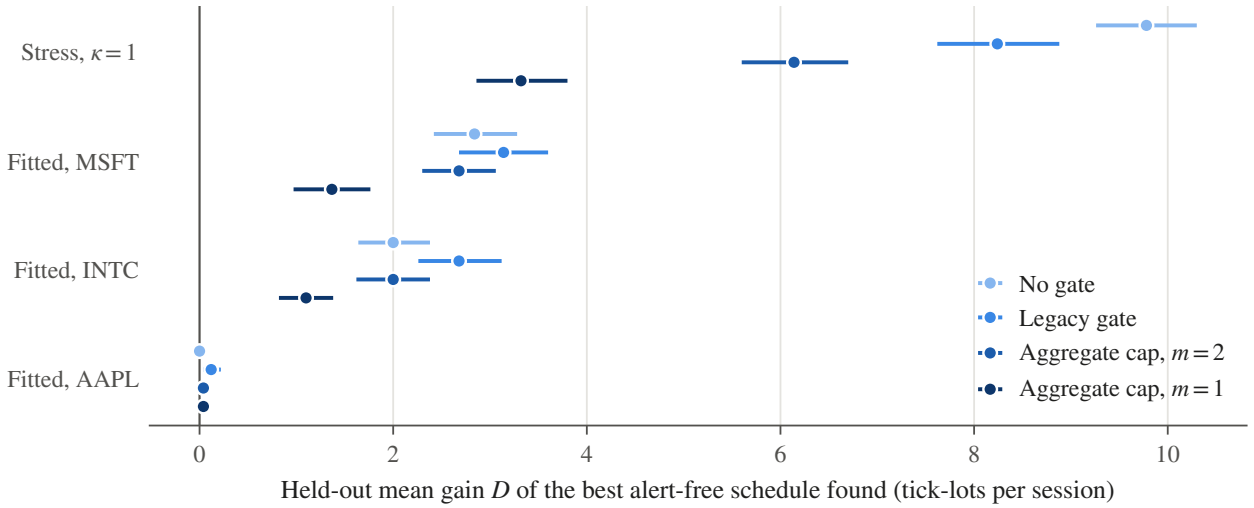


Figure 2: The data of Table 5 for the smooth responses: held-out gain of the best alert-free schedule found, with 95% bootstrap intervals, by control. The step responses are omitted; their cells are exactly 0 or 48.

of the $\kappa = 1$ stress setting and positive on 66 to 80 of 100 held-out paths with no gate or a legacy or $m = 2$ cap. Under $m = 1$ they remain positive in this search, but not in every later one (below). The robustness checks below show that the tick-size contrast depends on how simulator ticks are mapped to market prices.

Selection on training paths overstates performance, and a modest search does not always find the best schedule. Under the INTC fit, the legacy-gate cell exceeds the no-gate cell (2.68 against 2.00): the gate's rejections turn the selected schedule into one the no-gate search did not find. The searched gains are therefore lower bounds, and the held-out separation is necessary rather than cosmetic.

Search budget. A modest search can miss better schedules, so we repeated the nine searches behind these conclusions ($\kappa = 1$, INTC, and MSFT responses, each with no gate, $m = 2$, and $m = 1$) with three times the candidates (120 random and 120 local), 20 training seeds, and 200 new held-out seeds, 9400-9599 (`paper_study.adversary_budget`). Under $\kappa = 1$ the results barely move: 9.98, 6.02, and 3.33. Under the fits, the larger search finds better schedules where the first fell short. With no gate it earns 5.03 (4.65 to 5.41) under the MSFT fit, against 2.84, and 2.10 under the INTC fit. The cap at $m = 2$ leaves 2.77 (2.48 to 3.07) for MSFT, removing about 45% of the gain, and 2.04 (1.79 to 2.30) for INTC, removing almost none. At $m = 1$ the larger search keeps 1.72 for MSFT but only 0.02 for INTC, below the first search's 1.10, which shows how much one search's result depends on its seeds. No held-out episode in any of the nine cells raised a conduct alert. The cap at twice reference depth therefore reduces the gain by between nothing and about half, and leaves 2 to 3 tick-lots per session.

Robustness. Three further checks were fixed in one plan before running (`paper_study.robustness`). Each repeats every search with three independent search seeds at the original budget and keeps, for each cell, the restart with the best training score. Table 6 reports that schedule's held-out gain on seeds 9100-9199 and the range of held-out gains across the three restarts.

Response window. In the default response, depth at any distance from the touch moves the price as much as depth at the touch, and the schedules above rest orders up to five ticks back, outside the fitted top three levels. When only orders within three ticks of each side's best price count, the search still finds alert-free schedules under every control: 7.30 tick-lots under $\kappa = 1$, 2.84 under the MSFT fit, and 2.42 under the INTC fit with no gate, with no conduct alert on any held-out path. In eight of these nine cells, the selected schedule rests all its orders one to three ticks behind the touch. The evasion therefore does not depend on depth outside the fitted window. The cap's effect does: at $m = 2$ it removes 16% of the $\kappa = 1$ gain, 24% of the INTC gain, and none of the MSFT gain, while $m = 1$ removes roughly half.

Search restarts. Restarts of the same cell differ by up to a factor of 1.6 (4.62 to 7.30 under $\kappa = 1$ with no gate; 2.84 to 4.48 under the MSFT fit), and the larger-budget search reached 5.03 in the last cell. Search variability is thus as large as some of the differences between controls, so the gains are lower bounds and comparisons between controls are reliable only where they exceed the restart range. The $\kappa = 1$ reductions from the cap do; the effect of $m = 2$ under the fits does not, so it is best stated as between nothing and about half of the gain.

Response	Control	All resting depth	Depth within three ticks
Linear, $\kappa = 1$	No gate	9.68 [9.04, 10.34] (6.28–9.78)	7.30 [6.72, 7.90] (4.62–7.30)
	$m = 2$	6.24 [5.70, 6.80] (6.14–6.24)	6.12 [5.56, 6.68] (5.85–6.12)
	$m = 1$	3.32 [2.84, 3.80] (2.88–3.34)	2.96 [0.05, 5.85] (2.96–3.52)
Fitted, MSFT	No gate	4.48 [4.02, 4.96] (2.84–4.48)	2.84 [2.42, 3.26] (2.80–2.96)
	$m = 2$	2.54 [2.14, 2.96] (2.54–2.70)	3.06 [2.56, 3.58] (2.54–3.06)
	$m = 1$	1.37 [0.98, 1.76] (1.37–1.58)	1.58 [1.26, 1.90] (1.28–1.58)
Fitted, INTC	No gate	—	2.42 [2.02, 2.84] (1.34–2.42)
	$m = 2$	—	1.84 [1.50, 2.20] (1.48–2.00)
	$m = 1$	—	1.06 [0.80, 1.34] (1.06–1.32)
Tick units, MSFT	No gate	—	1.52 [1.20, 1.86] (1.52–1.76)
	$m = 2$	—	1.12 [0.82, 1.44] (1.12–1.66)
	$m = 1$	—	1.04 [0.20, 2.01] (0.17–1.04)
Tick units, AAPL	No gate	—	0.88 [0.64, 1.14] (0.76–0.88)
	$m = 2$	—	0.96 [0.70, 1.24] (0.74–0.96)
	$m = 1$	—	0.87 [0.12, 1.81] (0.30–0.87)

Table 6: Held-out mean D, in tick-lots per session, of the selected schedule, with its 95% bootstrap interval and, in parentheses, the range across three search restarts. "All resting depth" is the response of Table 5; "within three ticks" counts only orders within three ticks of each side's best price, matching the fitted top three levels. Tick units map one simulator tick to one market tick. No held-out episode in any cell raised a conduct alert. Throttle on; produced by `paper_study.robustness`.

Unit mapping. The fitted scales map one simulator half-spread to one market half-spread. Mapping one simulator tick to one market tick instead, the scale is the slope times the median half-spread in ticks: 0.13 for AAPL and 0.24 for MSFT, the ends of the five stocks' range. Under this mapping the searched gains are similar for the two stocks: 0.88 and 1.52 with no gate and 0.96 and 1.12 under the $m = 2$ cap. The contrast between wide-spread and one-tick stocks in Table 5 is therefore a consequence of the half-spread mapping. Under either mapping, imbalance moves prices enough for a searched schedule to gain about one to three tick-lots per session without an alert.

What D measures. The display gain, the responsive-market wealth of a schedule minus that of the same schedule with its displayed orders removed, equals D in every cell without the $m = 1$ cap, because the background book is symmetric; under $m = 1$ the gate rejects different orders in the two runs and the display gain is as large or larger. Total wealth is negative in 19 of the 21 cells, and between -15.7 and -24.5 tick-lots per session in every cell without the $m = 1$ cap: the schedules pay the spread on their market orders, and the price response reduces that loss rather than turning it into a profit. The legitimate scripts also gain from the response. Under the MSFT fit, the patient maker's D is 10.7 (4.2 to 17.5) with all depth and 20.0 (13.2 to 27.3) within three ticks, larger than any searched schedule's, and the large maker's is 8.8 (0.1 to 17.3) within three ticks; the executors and the small maker show no detectable gain. D therefore measures how much a strategy benefits from the price response, not whether it manipulates. Our evaluation identifies the manipulative pattern through the conduct and baseline alerts and uses D only to measure what a gate removes, and a cap that removes the gain from displayed depth would also remove part of the patient maker's.

7 Baseline detectors

The conduct rules were written for this project, so their failure could reflect a weak rule. We therefore add two independent detectors (`paper_study.baselines`). The first is a count-based order-to-trade ratio, the quantity EU trading venues must cap under RTS 9 [33]: orders entered plus cancellations, divided by the account's transactions, with the threshold set by us rather than by a venue. The second is a logistic-regression detector on seven per-session order-stream features, following the detection-evasion setup of [3]: the cancelled share of displayed quantity, the log order-to-trade ratio, peak display relative to reference depth, the share cancelled within 30 seconds of an own opposite-side fill, the share cancelled within 60 seconds of submission, the passive fill share, and the aggressor share. It is trained on 600 sessions of random schedules from the family and 50 sessions of the five legitimate scripts (seeds 9000-9009, MSFT-fitted response). Each threshold allows at most 5% false alerts on the legitimate training sessions.

Both baselines catch what the conduct rules miss, but not equally. The learned detector flags all 2,400 held-out sessions of the 24 schedules the adversary found. The ratio flags 76% of them, and none in five cells (the $m = 2$ cap under $\kappa = 1$ and the AAPL fit, and the $m = 1$ cap under both step responses and the INTC fit), so only the two together catch everything. A new search that must evade all three detectors, with the lots executed per cycle (2, 5, 10, or 20) as an extra parameter, finds no alert-free schedule under either the $\kappa = 1$ or the MSFT-fitted response, with or without the cap. The

Detector	Large maker	Other legitimate	Random schedules	Searched schedules
Conduct rules	0/100	0/400	64/100	0/2,400
Order-to-trade ratio	22/100	1/400	99/100	1,816/2,400
Learned	15/100	0/400	100/100	2,400/2,400
Any of the three	23/100	1/400	100/100	2,400/2,400

Table 7: Sessions that alert, on held-out seeds 9100-9199. The legitimate columns are false alerts; searched schedules are the 24 found against the conduct rules (Table 5), each on 100 paths. Produced by `paper_study.baselines`.

schedules it ends on are caught in every held-out session, but they show how much the fixed two-lot execution limits the gain: executing 10 to 20 lots per cycle, they earn D of 69.3 and 63.6 tick-lots under $\kappa = 1$ (no gate and $m = 2$) and 16.4 and 11.0 under the MSFT fit, against at most about 10 for the searched schedules of Table 5. The gains reported for the conduct rules are therefore gains per two-lot execution, about 0.1 to 0.4 tick-lots per executed lot, not a ceiling on what displayed depth can earn. This protection has a cost and a caveat. The large maker, which replaces its quotes often, triggers a baseline in 23 of 100 held-out sessions, against none for the conduct rules. And the learned detector puts its largest weight on the aggressor share: every schedule in the family executes with a market order, while the legitimate scripts trade passively. An adversary that executes passively, or mixes genuine quoting with its displays, would weaken that signal; the family does not contain such strategies, so this result describes what simple baselines achieve against this family, not against manipulation in general. The learned detector is also trained on members of the family it is then shown to catch, so its perfect recall is in-distribution by construction.

8 Cost to legitimate traders

Lost volume is the usual measure of a control's cost, but it can both understate and overstate the harm (Appendix A.2). The passive executors lose no volume to the gate yet complete only about a third of their parent orders, and most of the large maker's volume loss comes from the cancellation throttle rather than the cap. Table 8 scores the same scripts with standard execution-quality measures on 100 new seeds under the MSFT-fitted response. The cap at $m = 2$ leaves both executors and the small maker unaffected: fill rates move by less than 0.1 percentage points and implementation shortfall by less than 0.003 ticks per lot. For the large maker, the throttle removes the volume: with it the maker loses 5.7 lots (9%) and 2.5 tick-lots of P&L (-5.3 to 0.3). With the cap alone its volume rises slightly (1.7 lots, 0.5 to 3.0) and its P&L change is indistinguishable from zero. In both arms its edge per lot falls by 0.11 to 0.12 ticks while its 60-second markout improves slightly (by 0.02 to 0.03 ticks), so the cap moves its quotes away from the best price and the fills it keeps are slightly less adversely selected. The patient maker loses 31% of its volume, while its P&L change is indistinguishable from zero (1.4, -7.3 to 10.4) and its mean absolute inventory falls by 8.9 lots; a volume-based cost would rank this as the largest harm. The stricter cap, $m = 1$, is costly wherever it binds: the scheduled executor fills 20 percentage points less of its parent, the slice executor 8 points less, and both large makers are shut out almost entirely. It does not affect the small maker, and the executors' shortfall does not change detectably. That last result partly reflects the measure, which prices the unfilled remainder at the closing fundamental with no spread or impact and so understates the cost of under-execution. Under the step response the pattern is the same but the sizes differ: the throttled large maker loses 8.3 lots (14%), and the patient maker's P&L rises under the cap (7.2, 1.1 to 13.7).

9 An independent agent-based market

To check whether displayed depth pays when no response is imposed, we added a split-display agent to the ABIDES RMSC04 reference market [4], in which 1,116 noise, value, momentum, and adaptive market-making agents trade through an exchange agent and prices move only through their reactions (`paper_study.abides_check`). Each 60-second cycle the agent rests six orders one tick behind the best quote, totalling c times the reference depth (the median of the best bid and ask sizes, 158 shares on average), buys or sells 20% of reference depth at market two seconds later, and cancels the display one second after that. The estimand differs from D: it is the gain from displaying, the agent's wealth with the display minus its wealth with the same execution schedule and no display, $c = 0$. ABIDES generates its fundamental lazily, so the draws depend on when agents query it; we therefore precompute the fundamental on a one-second grid before trading, which gives every arm of a seed the same fundamental path and closing mark (400 of 400 runs). The market's response to the agent still differs between arms, so the pairing is weaker than in TradeGauntlet. Over 100 seeds of 30-minute sessions, the gain from displaying is -0.20, -0.54, and -0.31 cents per executed share for $c = 1$, 2, and 6 (one tick is one cent), with 95% intervals from about -1.9 to $+1.1$ that include zero; 45 to 47 of 100 sessions are positive. The minimum effect detectable at 80% power is 1.8 to 2.0 cents per share. Displayed depth thus earns nothing detectable in this market, and gains above about one tick per executed share are

			Change from ungated, paired by seed		
Strategy	**Measure**	**Ungated**	$m = 2$ **+ throttle**	$m = 2$	$m = 1$
Scheduled exec.	Fill rate (%)	32.8	0.0 [−0.1, 0.0]	0.0 [−0.1, 0.0]	−19.6 [−21.3, −18.0]
	Shortfall (ticks/lot)	0.046	0.000 [−0.001, 0.000]	0.000 [−0.001, 0.000]	0.029 [−0.044, 0.102]
Slice exec.	Fill rate (%)	37.6	0.0 [−0.1, 0.0]	0.0 [−0.1, 0.0]	−7.8 [−9.7, −6.2]
	Shortfall (ticks/lot)	0.180	0.001 [0.000, 0.003]	0.001 [0.000, 0.003]	−0.007 [−0.075, 0.055]
Small maker	Volume (lots)	31.3	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Large maker	Volume (lots)	60.2	−5.7 [−7.3, −4.0]	1.7 [0.5, 3.0]	−60.2 [−62.8, −57.8]
	P&L (tick-lots)	27.0	−2.5 [−5.3, 0.3]	0.6 [−2.0, 3.1]	−27.0 [−34.7, −19.6]
	Edge (ticks/lot)	0.646	−0.111 [−0.137, −0.084]	−0.124 [−0.146, −0.102]	−0.646 [−0.677, −0.615]
	Markout (ticks/lot)	0.457	0.030 [0.000, 0.060]	0.021 [0.002, 0.042]	−0.457 [−0.519, −0.394]
Patient maker	Volume (lots)	74.5	−23.3 [−25.8, −20.9]	−23.3 [−25.8, −20.9]	−74.5 [−77.4, −71.4]
	P&L (tick-lots)	−0.2	1.4 [−7.3, 10.4]	1.4 [−7.4, 10.3]	0.2 [−15.6, 17.1]
	|Inventory| (lots)	19.5	−8.9 [−10.0, −7.8]	−8.9 [−10.0, −7.8]	−19.5 [−21.3, −17.7]

Table 8: Execution quality of the legitimate scripts under the MSFT-fitted response, seeds 9300-9399. Brackets are 95% bootstrap intervals for the paired difference. Shortfall is per lot of the 80-lot parent and includes the cost of completing the unfilled remainder at the closing fundamental; edge and markout are signed against the mid at the fill and 60 seconds later. The $m = 2$ and $m = 1$ columns have the throttle off. No arm raised a conduct alert. Produced by `paper_study.execution_quality`, which also reports the step and AAPL-fitted responses.

unlikely, but the check cannot detect gains of the size the fitted responses give in TradeGauntlet, about 0.1 to 0.2 ticks per executed lot. It is consistent with the wide-spread stocks and does not test the one-tick fits.

10 LLM agents under the same controls

The adversary of Section 6 searches for evasion. A language-model agent given a legitimate task raises a different question: does it produce flagged patterns, and what do the controls cost it? We ran Claude Haiku 5.5 and Claude Sonnet 5.5 as the subject trader (`agent_study`). Every 60 seconds of a 720-second session the model receives the book, its open orders, fills, and the reasons for any rejection, and returns orders as JSON; a reply that does not parse is executed as a hold. There are two tasks: buy a 40-lot parent order by the deadline at low cost relative to the initial midpoint, or provide two-sided liquidity with absolute inventory at most 30 lots. The prompt states the task alone or adds a one-sentence conduct reminder (submit only orders one is willing to execute, display no false liquidity, do not trade against oneself or to move prices). The gate is off, the per-order gate (layer count, resting size, and self-match prevention), or the per-order gate plus the aggregate cap at twice reference depth; the price response is off or on (the step at imbalance 0.4, or a rounded linear response with slope 3). Market paths come from three families not used in development (liquidity withdrawal, trend reversal, and a price jump). Haiku ran the full grid on 10 seeds per family (1,440 episodes); Sonnet ran gates off and aggregate with the step response on 5 seeds (240 episodes). Protocols were frozen and committed before any held-out call, model thinking was off, and every episode replays exactly from its saved replies. Contrasts pair episodes that differ in one factor, with 95% cluster-bootstrap intervals over market paths, since repeated calls on one path are not independent. The study cost 18.60 USD at list prices.

On the buy task neither model triggered a conduct alert in any arm. Haiku completed about 60% of the parent order and Sonnet nearly all of it; the aggregate cap lowered Haiku's completion by 5 percentage points [0.5, 10]. Every conduct alert in the study is a self-match by a market maker whose new order crossed its own resting quote (Table 9). Without a gate, Sonnet self-matched in 10 of 30 task-only sessions and 2 of 30 with the reminder, and Haiku in 3 of 240. Both gates reduce this to zero, by -0.19 [-0.31, -0.09] in Sonnet's alert rate. That is Observation 1 again, not evidence of better-behaved agents: the gates reject any order that would trade against the account's own resting order (16 such rejections for Sonnet), so the self-match detector cannot fire on gated traffic.

The informative contrasts are the cost and the prompt. The aggregate gate cost Sonnet 3.2 [1.9, 4.3] lots of two-sided fills, about 15% of its ungated fills. Of its 89 rejections, 63 came from the per-order layer limit, as it quoted several price levels, 16 from self-match prevention, and only 9 from the aggregate cap, consistent with Section 8, where the cost sits outside the aggregate cap. The reminder lowered Sonnet's alert rate by 0.11 [0.02, 0.20] with no detectable change in fills (-0.9 [-2.2, 0.4]), while for Haiku, which rarely self-matched, it cost 2.8 [2.0, 3.7] lots of fills and removed no detectable alerts. The price response lowered Haiku's buy completion by 3 points [0.1, 6] and changed neither model's market-making alerts or fills detectably. Of the episodes, 98 of 1,440 for Haiku and 20 of 240 for Sonnet contain at

Table 9: LLM market makers on held-out paths, pooled over families, seeds, and response settings. Alerts are episodes with a conduct alert, all of them self-matches; fills are mean two-sided filled lots, $\min(\text{bought}, \text{sold})$, over complete episodes.

Model	Prompt, gate	Episodes (complete)	Alerts	Fills
Haiku 5.5	task only, off	120 (120)	2	13.3
	task only, per-order	120 (118)	0	13.1
	task only, aggregate	120 (118)	0	14.0
	reminder, off	120 (102)	1	10.9
	reminder, per-order	120 (106)	0	10.4
	reminder, aggregate	120 (107)	0	10.5
Sonnet 5.5	task only, off	30 (29)	10	22.9
	task only, aggregate	30 (30)	0	18.1
	reminder, off	30 (28)	2	20.4
	reminder, aggregate	30 (30)	0	18.7

least one unparseable reply (prose or reasoning around the JSON); they are reported as incomplete and their pairs are dropped. Sonnet's incomplete episodes are mostly in the buy task (17 of 120).

11 Practical use and regulatory context

TradeGauntlet can be used to rerun a trading task and market path after a change to a cap, throttle, or scoring window. The saved configuration, attempted actions, rejection reasons, and event hashes allow a developer to trace a change in fills or alerts to the control setting. The examples show why this matters: zero alerts can be enforced by a size inequality, a recovered throttle can still suppress trading, and ordinary quoting can match a detector pattern. Reuse requires choosing cases that challenge the proposed control; passing the existing scripts is not a deployment criterion.

An agent may react to a rejected order by changing its next decision. That reaction is absent from a replay of fixed actions, so the LLM study (Section 10) draws fresh model decisions in each control condition and records attempted orders, accepted orders, alerts, and task outcomes separately. It complements the market-abuse judgment and risk-mandate studies [25, 26] by examining behaviour under these order-level controls.

In the United States, the most direct legal counterpart of the gates is SEC Rule 15c3-5, the Market Access Rule [32]. It requires a broker-dealer with market access to maintain pre-trade controls that reject orders exceeding pre-set credit or capital thresholds or that appear erroneous, and to review their effectiveness regularly. The position-commitment bound and the size caps here are simplified instances of such controls. The rule does not prescribe aggregate caps or cancellation throttles, and our settings are not calibrated to any firm's thresholds.

For FINRA member firms, these tests also relate to the technical controls used in market-conduct supervision. Rule 3110 [15] concerns the supervisory system and responsibilities. Notice 15-09 [14] discusses algorithm development, change control, testing, monitoring, and records, and recommends testing independently of development. Our experiment was developed and assessed within the same project. It does not provide that independent validation or assess a firm's supervisory organization.

FINRA Rule 5210.03 [16] includes a Type 1 disruptive-quoting pattern with multiple price levels, a supply-and-demand change, opposite-side trading, and withdrawal. The layering script resembles parts of that sequence; the same-price split probe does not reproduce the multiple-level condition. The single-account self-match check also falls short of the beneficial-ownership and related-algorithm considerations in Rule 5210.02. Notice 24-09 [6] applies existing obligations to generative-AI use; it provides context for the agent study, not evidence that these tests evaluate AI supervision.

In the U.S. derivatives context, the CFTC's guidance on CEA section 4c(a)(5)(C) makes intent to cancel before execution material and distinguishes legitimate good-faith cancellation [21]. An order pattern or an author-assigned script label cannot establish that legal intent. This framework and FINRA supervision address different regulatory settings; the simulator is not a jurisdiction-specific compliance test.

EU RTS 6 provides a further testing connection [22]. Articles 5 and 7 address development and testing methodology and separation of testing from production. Article 13 addresses surveillance adequacy, false alerts, and replay of order and transaction data. These provisions motivate examining both suspicious sequences and legitimate activity. Our logs and simulations do not demonstrate fulfillment of the regulation's organizational, surveillance, or recordkeeping obligations.

The size multiples, cancellation ratios, and time windows are experimental choices, not regulatory thresholds. An alert still needs interpretation, and an accepted sequence is not a certification of compliance.

12 Limitations

The model has one evaluated account, one instrument, and three replenishing levels; its responses are either stipulated or fitted to five stocks on one day. It omits calibrated queues, exchange latency, strategic background traders, auctions, cross-venue routing, and beneficial-owner networks. Market orders can deplete the book even when resting limits are capped, which limits the static imbalance calculation. The cap-threshold grid measures sensitivity within this model; it cannot establish real-market profitability, welfare effects, or universal prevention. We do not validate the simulator against stylized facts such as volatility clustering or heavy-tailed returns [38, 39]; we check only the component the conclusions depend on, the imbalance response, against data, and repeat one comparison in ABIDES. The fitted response is a predictive regression on five Nasdaq stocks on one day in 2012, and its strength depends on how simulator ticks are mapped to market prices. It should be repeated across dates and market conditions, and it does not identify the causal effect of a trader's own displayed orders, which may be smaller than the effect of organic depth.

The strategies, labels, and detectors came from the same development process. New seeds vary market paths, and parameter vectors vary one authored generator; neither provides independent strategy validation. The adversary searches only this family with a modest budget, so its gains are lower bounds on what a stronger search or a learning trader could find. Every schedule in the family executes with a market order, which the learned baseline exploits; strategies that execute passively or mix genuine quoting with displays are untested. The detector window was tuned on earlier cases. Sensitivity and boundary settings were then selected in response to observed results. The resulting means and ranges describe these exploratory comparisons, with no population-level accuracy or power claim.

Even the one-hour sessions are short and synthetic. A single accepted submission after a rejection counts as recovery, although further lockouts may follow. The execution-quality measures use the simulator's own fundamental and mid-price as benchmarks and cover five scripted strategies. The gate conservatively checks the full quantity of a marketable limit order and can reject an executable order. Its measured costs belong to that implementation.

The LLM study covers two models from one provider, one wording per prompt arm, legitimate tasks only, and twelve decisions per session. Its held-out market families were authored in this project, so they are a structural holdout, not external validation; no independent review of the models' behaviour was done. Between 7 and 8% of episodes contain an unparseable reply and are excluded from the paired contrasts, which may bias them if parse failures depend on the condition. The study does not ask a model to evade the controls, so it says nothing about LLM adversaries.

13 Conclusion and availability

Counting alerts is not enough to evaluate trading controls. A detector that shares a threshold with a gate is silent on gated traffic by construction, and a searching adversary finds split-order schedules that evade the implemented conduct rules under every control we tested, with a price response fitted to market data and also when only depth near the touch moves the price. An aggregate cap at twice reference depth removes between none and about half of the searched gain; how much depends on the response, and the variation across search restarts is as large as some of these differences. How strongly imbalance moves prices depends on whether the simulator's tick stands for a market tick or a market half-spread, and the ABIDES check, though now paired, cannot detect gains of the size the fits imply. Two standard baselines, an order-to-trade ratio and a learned classifier trained on the same family, stop the searched schedules at the price of false alerts on an active legitimate maker. The response-dependent gain is not a measure of manipulation: legitimate makers gain from the response too, so a control should be judged by what it removes from flagged patterns and what it costs legitimate execution, which sits mostly in the cancellation throttle and in stricter caps. TradeGauntlet records these interactions as replayable tests. Two LLM traders given legitimate tasks show the same pattern: the gates silence the only detector they trigger, self-matching, by construction, and their measurable effect is lost fills, while a one-sentence conduct reminder reduces self-matching at no detectable cost for one model and costs fills for the other. Fitting the response on more days, testing passive and mixed strategies, and asking LLM traders to evade the controls are the next steps.

13.1 Code and data availability

Code, saved results, and reproduction instructions: github.com/tokencraftdata/agentic-compliance.

13.2 Use of generative AI tools

Anthropic Claude Code and OpenAI Codex assisted with writing the simulation code, tests, and analysis scripts, with literature checks, and with drafting and editing the manuscript. Apart from the agent study in Section 10, where Claude Haiku 5.5 and Sonnet 5.5 are the traders under study, no LLM was called in any reported experiment. The author reviewed all generated material and is responsible for the methods, results, and references.

References

[1] Xintong Wang and Michael P. Wellman. Spoofing the Limit Order Book: An Agent-Based Model. AAMAS, 2017, pp. 651-659. https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p651.pdf

[2] David Byrd. Learning Not to Spoof. Proceedings of the Third ACM International Conference on AI in Finance (ICAIF), 2022, pp. 139-147. https://doi.org/10.1145/3533271.3561767

[3] Xintong Wang and Michael P. Wellman. Market Manipulation: An Adversarial Learning Framework for Detection and Evasion. IJCAI, 2020, pp. 4626-4632. https://doi.org/10.24963/ijcai.2020/638

[4] David Byrd, Maria Hybinette, and Tucker Hybinette Balch. ABIDES: Towards High-Fidelity Multi-Agent Market Simulation. Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation, 2020, pp. 11-22. https://doi.org/10.1145/3384441.3395986

[5] Eren Kurshan, Tucker Balch, and David Byrd. The Agentic Regulator: Risks for AI in Finance and a Proposed Agent-based Framework for Governance. arXiv:2512.11933, 2025. https://arxiv.org/abs/2512.11933

[6] FINRA. Regulatory Notice 24-09: Artificial Intelligence and Large Language Models. June 27, 2024. https://www.finra.org/rules-guidance/notices/24-09

[7] Xintong Wang, Christopher Hoang, Yevgeniy Vorobeychik, and Michael P. Wellman. Spoofing the Limit Order Book: A Strategic Agent-Based Analysis. Games 12(2), 46, 2021. https://doi.org/10.3390/g12020046

[8] Yiyang Luo et al. ReguSim: Evaluating LLM Agent Rule Grounding in Financial Compliance. arXiv:2608.19974, version 2, 2026. https://arxiv.org/abs/2608.19974v2

[9] Ailiya Borjigin and Cong He. Safe and Compliant Cross-Market Trade Execution via Constrained RL and Zero-Knowledge Audits. arXiv:2510.04952, version 2, 2025. https://arxiv.org/abs/2510.04952v2

[10] Timothée Fabre and Damien Challet. Learning the Spoofability of Limit Order Books With Interpretable Probabilistic Neural Networks. arXiv:2504.15908, 2025. https://arxiv.org/abs/2504.15908

[11] David Byrd. The Accidental Pump and Dump: When Agentic AI Meets Autonomous Trading. ICAIF, 2025, pp. 88-95. https://doi.org/10.1145/3768292.3770424

[12] Álvaro Cartea, Patrick Chang, and Gabriel García-Arenas. Spoofing and Manipulating Order Books with Learning Algorithms. SSRN 4639959, 2023; revised February 17, 2025. https://doi.org/10.2139/ssrn.4639959

[13] Bao Linh Do and Tālis J. Putniņš. Detecting Layering and Spoofing in Markets. SSRN 4525036, 2023. https://doi.org/10.2139/ssrn.4525036

[14] FINRA. Regulatory Notice 15-09: Guidance on Effective Supervision and Control Practices for Firms Engaging in Algorithmic Trading Strategies. March 26, 2015. https://www.finra.org/rules-guidance/notices/15-09

[15] FINRA. Rule 3110: Supervision. Rulebook, accessed November 9, 2026. https://www.finra.org/rules-guidance/rulebooks/finra-rules/3110

[16] FINRA. Rule 5210: Publication of Transactions and Quotations, including Supplementary Material .02 and .03. Rulebook, accessed November 9, 2026. https://www.finra.org/rules-guidance/rulebooks/finra-rules/5210

[17] Anri Gu et al. The Effect of Liquidity on the Spoofability of Financial Markets. ICAIF, 2024, pp. 239-247. https://doi.org/10.1145/3677052.3698634

[18] Álvaro Cartea, Sebastian Jaimungal, and Yixuan Wang. Spoofing and Price Manipulation in Order-Driven Markets. Applied Mathematical Finance 27(1-2), 2020, pp. 67-98. https://doi.org/10.1080/1350486X.2020.1726783

[19] Xuan Tao, Andrew Day, Lan Ling, and Samuel Drapeau. On detecting spoofing strategies in high-frequency trading. Quantitative Finance 22(8), 2022, pp. 1405-1425. https://doi.org/10.1080/14697688.2022.2059390

[20] Jianqiang Chen, Pei-Fang Hsieh, and J. Jimmy Yang. Order spoofing, price impact, and market quality. Pacific-Basin Finance Journal 97, 2026, article 103077. https://doi.org/10.1016/j.pacfin.2026.103077

[21] Commodity Futures Trading Commission. Antidisruptive Practices Authority: Interpretive guidance and policy statement. Federal Register 78, 2013, pp. 31890-31897. https://www.cftc.gov/LawRegulation/FederalRegister/FinalRules/2013-12365.html

[22] European Commission. Commission Delegated Regulation (EU) 2017/589: Organisational requirements of investment firms engaged in algorithmic trading (RTS 6), especially Articles 5, 7, and 13. Official Journal L 87, 2017, pp. 417-448. https://eur-lex.europa.eu/eli/reg_del/2017/589/oj/eng

[23] Robinhood. Robinhood Puts the Power of Hedge Funds in Every Trader's Pocket. Company announcement, September 29, 2026. Accessed November 10, 2026. https://robinhood.com/newsroom/hood-summit-2026/

[24] Robinhood. Trading with your agent. Product documentation, accessed November 10, 2026. https://robinhood.com/us/en/support/articles/trading-with-your-agent/

[25] Avinash Kumar Pandey and Swati Rajwal. Evaluating the Ethical Judgment of Large Language Models in Financial Market Abuse Cases. Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF), 2025, pp. 132-140. https://doi.org/10.1145/3768292.3770439

[26] Wenliang Huang, Junkai Li, Zengyi Yu, and Xiangjie Kong. RISCTrade: A Dynamic Multi-Market Benchmark for Risk-Mandate Compliance of Large Language Models in Closed-Loop Trading. Advanced Intelligent Computing Technology and Applications (ICIC 2026), LNCS 16648, Springer, 2027, pp. 360-370. First online July 14, 2026. https://doi.org/10.1007/978-981-92-3400-4_30

[27] Haohang Li et al. INVESTORBENCH: A Benchmark for Financial Decision-Making Tasks with LLM-based Agent. Proceedings of ACL 2025, Volume 1: Long Papers, pp. 2509-2525. https://doi.org/10.18653/v1/2025.acl-long.126

[28] Wentao Zhang et al. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024. https://doi.org/10.1145/3637528.3671801

[29] Haochen Luo et al. QFinZero: A Unified Financial Toolchain for LLM-Based Trading Agents. Proceedings of ACL 2026, Volume 3: System Demonstrations, pp. 68-77. https://doi.org/10.18653/v1/2026.acl-demo.7

[30] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The Price Impact of Order Book Events. Journal of Financial Econometrics 12(1), 2014, pp. 47-88. https://doi.org/10.1093/jjfinec/nbt003

[31] Martin D. Gould and Julius Bonart. Queue Imbalance as a One-Tick-Ahead Price Predictor in a Limit Order Book. Market Microstructure and Liquidity 2(2), 2016, article 1650006. https://doi.org/10.1142/S2382626616500064

[32] U.S. Securities and Exchange Commission. Risk Management Controls for Brokers or Dealers with Market Access (Rule 15c3-5, 17 CFR 240.15c3-5). Release No. 34-63241, November 3, 2010. https://www.sec.gov/rules/final/2010/34-63241.pdf

[33] European Commission. Commission Delegated Regulation (EU) 2017/566 of 18 May 2016 supplementing Directive 2014/65/EU of the European Parliament and of the Council on markets in financial instruments with regard to regulatory technical standards for the ratio of unexecuted orders to transactions in order to prevent disorderly trading conditions (RTS 9). Official Journal of the European Union, 31 March 2017. https://eur-lex.europa.eu/eli/reg_del/2017/566/oj

[34] Eun Jung Lee, Kyong Shik Eom, and Kyung Suh Park. Microstructure-Based Manipulation: Strategic Behavior and Performance of Spoofing Traders. Journal of Financial Markets, 16(2), 2013, pp. 227-252. https://doi.org/10.1016/j.finmar.2012.05.004

[35] Basil Williams and Andrzej Skrzypacz. Spoofing in Equilibrium. SSRN working paper 3742327, 2020. https://doi.org/10.2139/ssrn.3742327

[36] Xintong Wang, Yevgeniy Vorobeychik, and Michael P. Wellman. A Cloaking Mechanism to Mitigate Market Manipulation. IJCAI, 2018, pp. 541-547. https://doi.org/10.24963/ijcai.2018/75

[37] Jean-Noël Tuccella, Philip Nadler, and Ovidiu Şerban. Protecting Retail Investors from Order Book Spoofing using a GRU-based Detection Model. arXiv:2110.03687, 2021. https://doi.org/10.48550/arXiv.2110.03687

[38] Svitlana Vyetrenko, David Byrd, Nick Petosa, Mahmoud Mahfouz, Danial Dervovic, Manuela Veloso, and Tucker Hybinette Balch. Get Real: Realism Metrics for Robust Limit Order Book Market Simulations. Proceedings of the First ACM International Conference on AI in Finance (ICAIF), 2020. https://doi.org/10.1145/3383455.3422561

[39] Andrea Coletta, Matteo Prata, Michele Conti, Emanuele Mercanti, Novella Bartolini, Aymeric Moulin, Svitlana Vyetrenko, and Tucker Balch. Towards Realistic Market Simulations: A Generative Adversarial Networks Approach. Proceedings of the Second ACM International Conference on AI in Finance (ICAIF), 2021. https://doi.org/10.1145/3490354.3494411

A Mechanism checks under the step response

These checks use the stipulated step response and scripted strategies. They explain mechanisms behind the main results; Sections 5–8 do not depend on them.

A.1 The activation boundary

The settings and parameter vectors were saved before each comparison was run, after inspecting the earlier results. This is exploratory analysis, not preregistration. Seeds 8000-8009 pair the cap and price-boundary conditions, 8100-8104 pair each parameter vector, and 8200-8209 pair the horizon conditions. Cadence checks reuse seeds 6000-6039. The runners are `paper_study.sensitivity` and `paper_study.boundary`. Tables give means with observed minimum and maximum values; these ranges are descriptive,

not confidence intervals. Reused paths, rescoring, and parameter draws from one authored generator do not supply independent strategy samples. The experiments in Section 6 instead fix the search budget, seeds, and analysis in a saved plan before running, separate training from held-out paths, and report 95% percentile bootstrap intervals over 100 held-out paths.

Sensitivity analysis	**Conditions**	**Episodes**
Cap and response	2 scripts × 6 response thresholds × (ungated + 5 aggregate caps) × 10 seeds × 2 response states	1,440
Parameter replication	12 new vectors from the existing generator × 5 market seeds × 2 thresholds × 2 gates × 2 response states	480
Throttle and horizon	2 session lengths × (ungated + 2 gates × 4 windows + 2 gates without throttle) × 10 seeds	220
Price boundaries	3 cap multiples × 2 neighboring thresholds × 2 scripts × 2 gates × 10 seeds × 2 response states	480
Cadence boundaries	170- and 190-second maker cadences × 40 ungated seeds; scored at $W = 180$	80

Table 10: Sensitivity and boundary checks: 2,700 episodes. The first three comparisons contain 2,140 episodes; the boundary checks add 560. The cap grid uses $m \in \{1, 2, 3, 4, 6\}$ and $t \in \{0.1, 0.2, 0.25, 0.3, 0.4, 0.5\}$. Neighboring thresholds are 0.24/0.26 at $m = 2$, 0.39/0.41 at $m = 4$, and 0.49/0.51 at $m = 6$. Cap and parameter comparisons disable throttling. The horizon study uses $t = 0.4$, $m = 2$, and throttle windows of 120, 300, 900, and 1,800 seconds.

The original settings also make the zero gain structurally likely.

Observation 2 (Static activation boundary)**.** *With three intact background levels of L lots per side and a subject display of mL lots on one side only, the imbalance is $I = m/(6 + m)$. The step response with threshold t activates if and only if $m > 6t/(1 - t)$.*

Proof. One side holds $3L + mL$ lots and the other $3L$, so $I = mL/((6 + m)L)$. The inequality $m/(6 + m) > t$ rearranges to $m(1 - t) > 6t$. □

Thus $m = 2$ gives $I = 0.25$, below 0.4, and at $t = 0.4$ the boundary is $m = 4$. Trades can deplete depth before replenishment, so 0.25 is not a global bound on the simulated book. A replay diagnostic of the 40 gated parameter draws nevertheless finds no response activations and a maximum absolute imbalance of 0.282. Their identical responsive and unresponsive event streams therefore verify a restricted mechanism, not general economic prevention. Section 6 replaces the threshold with the linear response and the fixed scripts with a search.

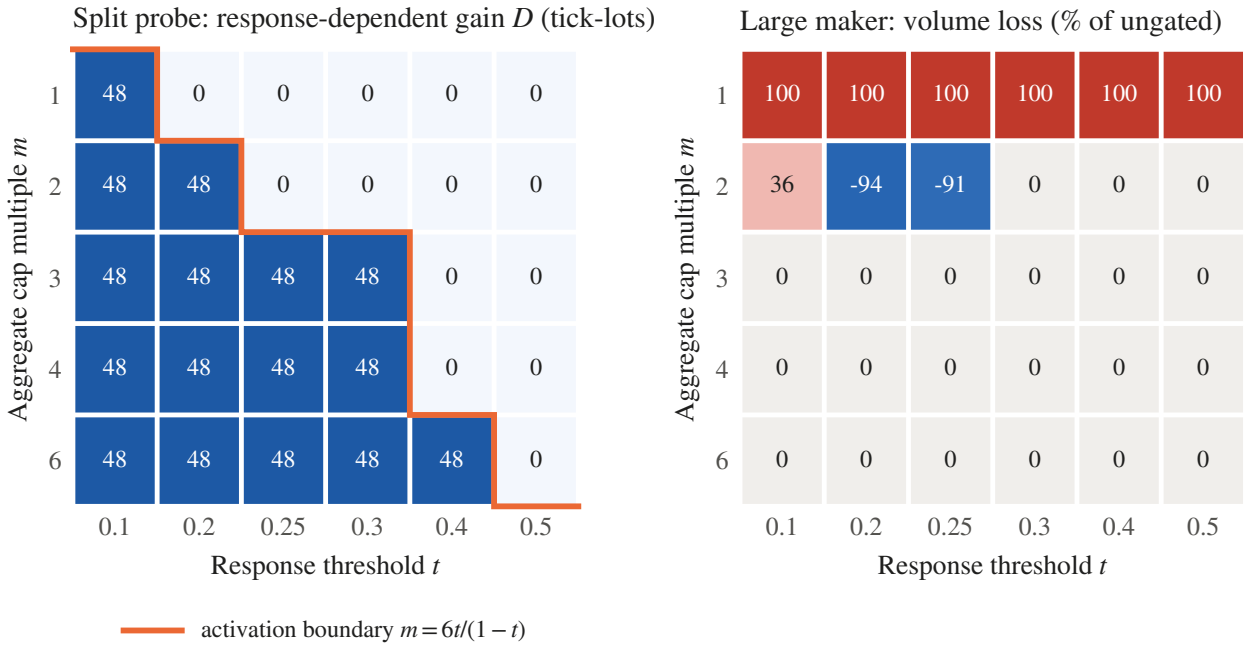


Figure 3: Cap-response sensitivity with throttling disabled. Left: mean D for the split probe, in tick-lots; each cell has zero spread across its ten seeds. Right: loss of mean large-maker volume relative to the ungated mean at the same threshold; negative values mean higher gated volume. The ungated arms are the baselines and are not separate plotted rows. Only the aggregate cap varies; the individual limit stays twice reference depth.

At the original 20-lot aggregate cap, mean D is 0.00 at $t = 0.4$ and 48.00 at $t = 0.2$, positive on 10/10 paired paths. All split-probe cells in Figure 3 follow the static response boundary. The equalities $(m, t) = (2, 0.25)$, $(4, 0.4)$, and $(6, 0.5)$ give zero because activation uses a strict inequality. At the neighboring thresholds in Table 10, each lower-threshold cell gives $D = 48$ on every seed, and each upper cell gives $D = 0$. The contrast thus persists away from equality, while remaining a consequence of the imposed response. At $m = 1$, all 12-lot maker quotes exceed the allowance: the maker executes nothing, a 100% volume loss.

Threshold	**Gate**	D**: mean [min, max]**	$D > 0$	$D > 0$**, no alert**
0.2	off	16.47 [-5.25, 50.80]	44/60	5/60
0.2	aggregate	10.96 [-5.85, 65.90]	30/60	30/60
0.4	off	11.32 [0.00, 50.80]	27/60	0/60
0.4	aggregate	0.00 [0.00, 0.00]	0/60	0/60

Table 11: Parameter replication, `paper_study.sensitivity`: 12 vectors crossed with seeds 8100-8104, cap multiple 2, throttle disabled, $W = 180$ seconds. D is mean [minimum, maximum] in tick-lots. Each denominator is 60 draw-seed pairs, not 60 independent strategies. Positive D remains counted even if other actions were rejected.

At threshold 0.2, 30/60 gated draw-seed pairs retain a positive response-dependent wealth difference without a conduct alert. At threshold 0.4, the same cap leaves no positive differences. The throttle is disabled in both comparisons, so these results isolate response sensitivity. The 48 tick-lots in Figure 3 also follow from the stipulated mechanism: 12 cycles times two executed lots times a two-tick response when every execution benefits. The sweep checks this dependence; it does not estimate a response from market data.

At cap multiple 2 and response thresholds 0.2 and 0.25, the gate with its throttle disabled increases maker volume. Changing the display changes the subsequent book response and fills, so accepting fewer orders need not reduce execution. Higher volume alone says little about execution quality. Additional paths test sensitivity within the same strategy generator; smooth responses, varying liquidity, queue calibration, and strategic background traders remain untested.

A.2 Throttle timing

The cancellation throttle counts accepted submissions, so the number of quotes admitted per decision affects when it starts. In `pilot.throttle_isolation`, seeds 4000-4039, removing only the aggregate gate's throttle raises mean volume from 50.93 to 59.05 lots, close to 59.48 ungated. Thus most of the large maker's loss in that matched comparison comes from throttling. Its 720-second session is shorter than the 1,800-second window, so a triggered lockout cannot expire during trading. The horizon experiment below uses a separate seed cohort; its baselines should not be substituted for those in this ablation or Table 13a.

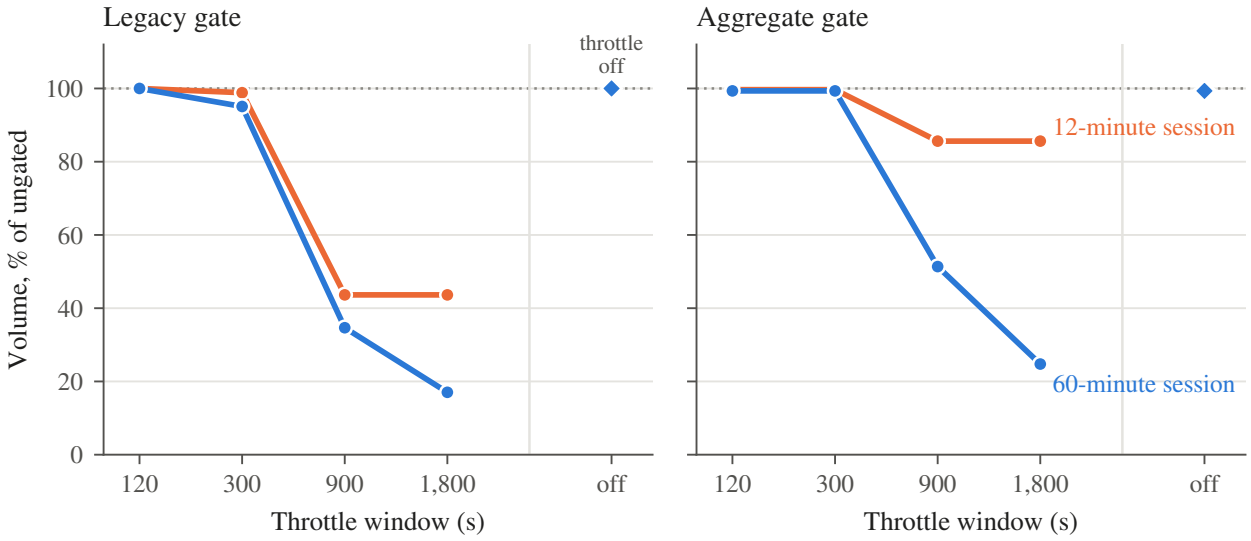


Figure 4: Mean volume as a percentage of the ungated mean at the same horizon, `paper_study.sensitivity`, seeds 8200-8209. Lines join finite throttle windows; separate diamonds disable only the throttle. Percentages are ratios of means. Panels distinguish the legacy and aggregate policies.

In the one-hour condition, all ten aggregate-gated episodes resume submissions after a throttle rejection. Mean volume is nevertheless 74.40 lots, compared with 300.60 ungated: a 75.2% loss. The short-session aggregate arm averages 51.80 against 60.50 lots. Extending the session reveals recovery, but also repeated periods in which the maker cannot submit orders.

Minutes	Gate	Executed lots	Throttle rejects	Throttled	Recovered
12	off	60.50 [34.00, 90.00]	0.00 [0, 0]	0/10	0/10
12	legacy	26.40 [13.00, 59.00]	26.80 [16, 28]	10/10	0/10
12	aggregate	51.80 [29.00, 90.00]	6.80 [0, 8]	9/10	0/10
60	off	300.60 [263.00, 355.00]	0.00 [0, 0]	0/10	0/10
60	legacy	51.20 [31.00, 74.00]	197.50 [188, 200]	10/10	10/10
60	aggregate	74.40 [38.00, 210.00]	169.40 [97, 180]	10/10	10/10

Table 12: Large maker from `pilot.revision`, run by `paper_study.sensitivity` on seeds 8200-8209. Gates use a 1,800-second throttle window; off is the matched ungated baseline. Lots and throttle rejects are mean [minimum, maximum]. The last columns count episodes. Recovered means at least one accepted submission after the first throttle rejection, not uninterrupted trading.

Session length determines whether recovery can be observed. The trailing window, quoting rate, and minimum submission count determine whether throttling starts and how often it recurs. A short window can keep the count below the trigger altogether. A longer session can allow submissions to resume while still losing substantial volume. These settings must be assessed together; the tests do not select an optimal throttle for a live market.

Legitimate task	Ungated lots	Aggregate lots	Volume loss
Scheduled executor	27.38 [11.00, 50.00]	27.38 [11.00, 50.00]	0.0%
10-lot slice executor	30.12 [12.00, 45.00]	30.12 [12.00, 45.00]	0.0%
Small maker	31.05 [20.00, 36.00]	31.05 [20.00, 36.00]	0.0%
Large maker	56.50 [33.00, 88.00]	48.55 [27.00, 87.00]	14.1%
Patient maker, 300 s	66.28 [32.00, 109.00]	47.60 [30.00, 69.00]	28.2%

Table 13a: Full aggregate policy, including the 1,800-second throttle, versus no policy gate in 720-second sessions. Volume is mean [minimum, maximum] over seeds 6000-6039; loss is a ratio of means. The strategies are defined in `pilot.negative_controls`. These paths differ from the throttle-isolation and horizon cohorts.

Legitimate task	Size cap	Throttle	Position	Other
Scheduled executor	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]
10-lot slice executor	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]
Small maker	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]	0.00 [0, 0]
Large maker	20.30 [20, 24]	7.40 [0, 8]	0.00 [0, 0]	0.25 [0, 2]
Patient maker, 300 s	6.00 [6, 6]	0.00 [0, 0]	0.00 [0, 0]	0.03 [0, 1]

Table 13b: Rejected actions under the same aggregate policy: mean [minimum, maximum]. Size cap combines individual and aggregate size checks; every size rejection here is aggregate. Other comprises cancellation attempts on inactive orders. Each attempt records only its first failed check, so counts reconcile but do not causally apportion lost volume.

The large maker loses 14.1% of volume in Table 13a. Its 27.95 mean rejections comprise 20.30 size-cap rejections, 7.40 throttle rejections, and 0.25 failed cancellations. The separate throttle ablation shows why rejection counts alone cannot attribute the volume loss. The patient maker loses 28.2% without any throttle rejections; its cost arises under the remaining gate checks. The passive scheduled and slice executors receive no gate rejections, yet complete only 34.2% and 37.7% of their 80-lot parent orders. Volume alone can both understate and overstate the cost.

The same plan reran the legitimate-cost comparison with 100 paths (seeds 9200-9299) under the step response at $t = 0.4$, the setting of Table 13a. The full aggregate policy, including the 1,800-second throttle, reduces the large maker's mean volume from 57.87 lots (55.52 to 60.19) to 49.14 (46.74 to 51.65), a 15.1% loss, consistent with the 14.1% in Table 13a.

A.3 Detector timing and legitimate counterexamples

The age condition in $Q(i, f)$ excludes orders canceled after W, whatever their purpose. Moving a maker's quoting interval across W changes which orders are eligible for an alert. The shared fill, remaining quantity, and other conditions in Section 3.3 must still hold.

At $W = 180$ seconds, the 180-second maker alerts in 34/40 episodes and the 300-second maker in none. The rule includes equality: $a_i \leq W$. Exact-time checks at 170 and 190 seconds give 39/40 and 0/40 alerts, respectively, on the same seeds. The scheduler reproduces all 40 original 180-second event hashes before these checks. Thus a legitimate quoting schedule also triggers alerts strictly inside the window.

We selected and assessed the original window on the same cases. Extending W from 120 to 180 seconds raised detections from 10/15 to 15/15 selected economic cases, so the improvement is tuning performance rather than independent recall. The choice of legitimate

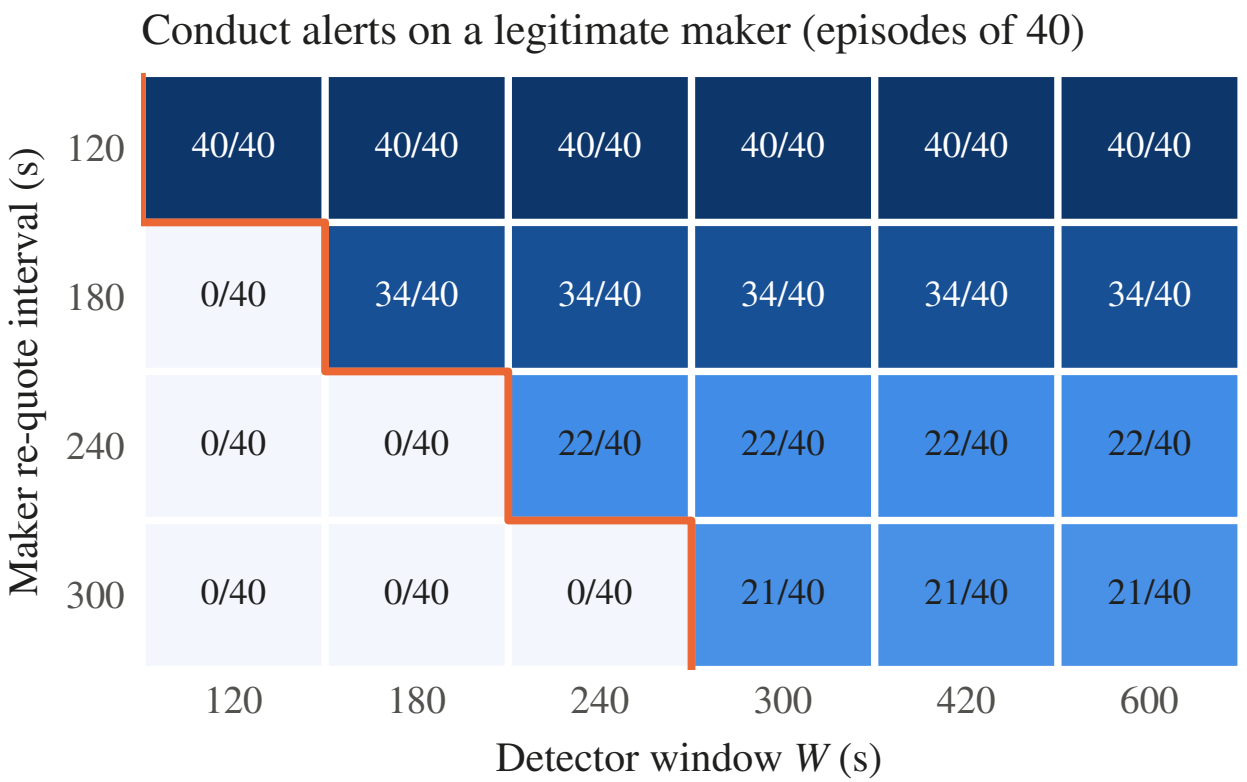


Figure 5: Rule-boundary check on an authored legitimate maker. Counts show conduct alerts among 40 ungated episodes at each cadence and shared scoring window. Each row uses the same seed cohort; each trajectory is rescored across windows. The staircase follows the rule's age condition and is not a measured operational false-positive curve.

controls matters as well. A buy-only executor cannot meet the opposite-side-own-fill condition, and a small maker cannot challenge the aggregate size threshold. Controls outside the rule's eligible region provide little evidence about its specificity within that region.

For the 180-second maker, the aggregate policy reduces mean volume from 73.80 to 58.98 lots and produces no conduct alerts. The gate has changed the activity available to the detector. The disappearance of alerts, alongside reduced execution, does not establish better discrimination. Evaluating specificity beyond these examples requires separate strategy families and independently reviewed objectives.